

Project Synopsis/Project Concept Document

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Description

Fleetix is an open-source route optimization system focused to solve complex employee transportation challenges. The system addresses two primary routing scenarios: transporting multiple employees from different locations to a single destination, and managing more complex many-to-many transportation routes. By leveraging Google APIs for geocoding and routing along with advanced optimization techniques, the solution aims to create efficient, flexible transportation networks that can handle multiple pickup and drop-off points while considering various operational constraints.

Profile of Users

Fleet Administrators

- Manage vehicle assignments and track transportation resources
- Require detailed insights into route efficiency and vehicle utilization
- Moderate to high technical skills

Employees

- End-users of the transportation service
- Varying levels of technical literacy
- Need simple, intuitive interface for viewing routes and schedules

Vehicle Drivers

- Need access to optimized routes
- Require mobile-friendly interface for navigation
- Rely on system for precise pickup and drop-off information

Usage Model and Diagrams

The system will be used through a web-based interface allowing users to input employee locations, define transportation parameters, and visualize optimized routes. Fleet managers can schedule and allocate vehicles, and employees can view their assigned pickup and drop-off details. The system will calculate routes considering factors like vehicle capacity,

time windows, and road networks using Google APIs for geocoding and routing services.

